

1. Escape, Chik Pan Wong & Aidan Anderson, 02, Spring 2026
2. **at least one image (screenshot) showing a typical scene from your game being played**



3. **instructions for compiling and running your game, including the network server**

To compile: run compile.bat

To run the server: run runServer.bat

To run the game: run run.bat

IP and ports can be set in each batch file

4. **how to play your game, including what things happen and how the scoring works**

- The player is trapped on a planet and explore the environment, find a key, and the spaceship to escape.

5. **Controls:**

- **Keyboard:**
 - W/S – move forward/backward
 - A/D – move left/right
 - E – take
 - Space – jump
 - Q – Shoot bullet
 - F – push
 - R – on/off flashlight
 - Z – camera zoom in

- X – camera zoom out
- C – camera left
- V – camera right
- B – camera down
- G – camera up
- O – enable rendering terrain
- P – disable rendering terrain
- **PlayStation gamepad:**
 - RX -- Azimuth Action
 - RY -- Elevation Action
 - Z -- Radius Action
 - B0 – Take
 - B1 – jump
 - B2 – Push
 - B3 – on/off flashlight
 - B6 – Shoot bullet

6. a description of the lighting used in your game, including any lights that turn on/off

- spotlight on/off – player’s flashlight
- Red positional light – key’s light
- Green positional light – player’s light
- spotlight – spaceship light

7. a brief summary of any changes (or none) that you made to the network protocol

No changes made to the network protocol.

8. a list of changes and additions that you made to TAGE

- CameraOrbit3D
- GameObject – yaw

9. a statement indicating the (1) genre, (2) theme, (3) dimensionality, and (4) activities utilized in your game (see week 1 notes -- chapter 00 -- for examples)

- **genre** – adventure
- **theme** – sci-fi planet escape
- **dimensionality** – 3D

- **Activity**
 - Exploration – Player moves around the planet to find the key and.
 - Collection – Player collects the key needed to activate the spaceship.
 - Enemy avoidance – Player should avoid enemies that chase and attack.
 - Goal – Player wins by reaching the spaceship with the key and escaping.

10. an explanation of where (in the game, not the code) each project requirement is visible

- **External Models:** Spaceship, enemy and dolphin models
- **Networked Multiplayer:** Game automatically connects to server if available
- **Skybox and terrain:** Stars background skybox and mars terrain visible in game
- **Lights:** Spotlight by spaceship, red positional light by key, green positional light that follows player, toggleable flashlight
- **HUD:** HUD is displayed in game
- **3D Sound:** Player plays footstep sound while walking, spaceship plays beep sound periodically.
- **Hierarchical Scenegraph:** Player, Bullet and Enemy Game Objects children of root object
- **NPCs:** Enemies follow and attack players using behavior tree logic
- **Physics:** Players can jump and move using their rigid body, bullets have a physics sphere that follows their trajectory

11. a list of the requirements that you weren't able to get working

- Did not complete player model
- Did not complete bullet-enemy hit detection
- Enemies do not spawn in single player (they do in multiplayer)

12. any technique you used in your game that goes beyond the requirements

13. contributions:

- Chik:
 - Skybox and Terrain
 - Hierarchical Scene Graph
 - Light

- HUD
- Animation
- NPC
- enemy model
- Aidan
 - Networking
 - Physics
 - Lighting
 - HUD
 - NPCs
 - Sound

14. a list of assets that you created yourself (models, textures, heightmap, etc.), and items obtained that were distributed in CSc-155 or CSc-165

- animation
 - attack.rka
 - enemy.rkm
 - enemy.rk
 - idle.rka
 - walk.rka
- model
 - enemy.obj
- textures
 - enemyUV.jpg

15. Source and evidence of permission for any item (models, textures, etc.)

Sources:

- skybox (galaxy): <https://tools.wwwtiro.net/space-3d/index.html#animationSpeed=1&fov=80&nebulae=true&pointStars=true&resolution=1024&seed=15iw8a3c0j6o&stars=true&sun=true>
license: <https://github.com/wwwtiro/space-3d?tab=Unlicense-1-ov-file#readme>
- spaceship: <https://sketchfab.com/3d-models/spaceship-colaid1-50k-34acc30c89364a668ab1025686a686bc>
license: <https://creativecommons.org/licenses/by/4.0/>

- heightmap: <https://commons.wikimedia.org/wiki/File:Heightmap.png>

license: same website

Licensing [edit]



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In some countries this may not be legally possible; if so:

*I grant anyone the right to use this work **for any purpose**, without any conditions, unless such conditions are required by law.*

- heightmap texture (floor.jpg): https://polyhaven.com/a/rock_face
license: CC0 <https://polyhaven.com/license>
- Foot step Sound: <https://opengameart.org/content/foot-walking-step-sounds-on-stone-water-snow-wood-and-dirt>
license: <https://www.gnu.org/licenses/gpl-3.0.html>
- Beep Sound: <https://opengameart.org/content/3-ping-pong-sounds-8-bit-style>
license: <https://creativecommons.org/publicdomain/zero/1.0/>
- key: <https://sketchfab.com/3d-models/crystal-key-473d184560c74490aeffefff8cfff23c>

license type: CC-BY-4.0 (<http://creativecommons.org/licenses/by/4.0/>)

16. which RVR-5029 lab machines (at least two – it's networked!) on which your program was tested and is known to work correctly on

Tested on ECS-PONG and ECS-WARCRAFT